

Correlated random effects: iidkd

This model is available for dimensions $k = 2$, to 24. We describe in detail the case for $k = 3$ as other ones are similar. This model do the same as models `iid2d`, `iid3d`, `iid4d`, `iid5d`, but uses a different and more efficient parameterisation.

Parametrization

The $(k = 3)$ -dimensional Normal-Wishard model is used if one want to define three vectors of “random effects”, u and v and w , say, for which (u_i, v_i, w_i) are iid bivariate Normals

$$\begin{pmatrix} u_i \\ v_i \\ w_i \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \mathbf{W}^{-1})$$

where the covariance matrix $\mathbf{W}^{-1}$ is parameterised as $\mathbf{W} = \mathbf{L}\mathbf{L}^T$, where

$$\mathbf{L} = \begin{pmatrix} \exp(\theta_1) & & \\ \theta_4 & \exp(\theta_2) & \\ \theta_5 & \theta_6 & \exp(\theta_3) \end{pmatrix} \quad (1)$$

and $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$ can take any value. The number of hyperparameters are $k(k+1)/2$, which is 3, 6, 10, 15, 21, 28, 36, 45, 55, ..., for $k = 2, 3, 4, 5, 6, 7, 8, 9, 10, \dots$

For these models the precision matrix $\mathbf{W}$ is Wishart distributed

$$\mathbf{W} \sim \text{Wishart}_k(r, \mathbf{R}^{-1}),$$

with density

$$\pi(\mathbf{W}) = c^{-1} |\mathbf{W}|^{(r-(k+1))/2} \exp \left\{ -\frac{1}{2} \text{Trace}(\mathbf{W}\mathbf{R}) \right\}, \quad r > k+1$$

and

$$c = 2^{(rk)/2} |\mathbf{R}|^{-r/2} \pi^{(k(k-1))/4} \prod_{j=1}^k \Gamma((r+1-j)/2).$$

Then,

$$\mathbb{E}(\mathbf{W}) = r\mathbf{R}^{-1}, \quad \text{and} \quad \mathbb{E}(\mathbf{W}^{-1}) = \mathbf{R}/(r - (k+1)).$$

Hyperparameters

The hyperparameters are $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$.

The prior-parameters are

$$(r, R_1, R_2, R_3, R_4, R_5, R_6)$$

where

$$\mathbf{R} = \begin{pmatrix} R_1 & R_4 & R_5 \\ R_4 & R_2 & R_6 \\ R_5 & R_6 & R_3 \end{pmatrix}$$

The `inla` function reports posterior distribution for the hyperparameters $\{\theta_i\}$, and the conversion into interpretable quantities can be done using simulation as described below.

The prior for θ is **fixed** to be `wishartkd`, and number of prior parameters required are $1 + k(k+1)/2$. By default the prior-parameters are

$$(r = 100, \underbrace{1, \dots, 1}_{k \text{ times}}, 0, \dots, 0)$$

The model `iidkd` is specified as

where $\text{order} = k = 3$, and the `iidkd` model is represented internally as one vector of length n ,

where $n = 3m$, and n is the (required) argument in $\mathbf{f}()$.

$$\sum u_i = 0, \quad \sum v_i = 0 \quad \text{and} \quad \sum w_i = 0.$$

(**Note:** The value “1048576” is just a code for “replace this by the default value”. As the default value depends on **order**, this was the easy way out for the moment.)

hyper

```
hyperid 29101  
name theta1  
short.name theta1  
initial 1048576  
fixed FALSE  
prior wishartkd  
param 30 1048576 1048576 1048576 1048576 1048576 1048576 1048576 1048576 1048576 1048576 1048576  
to.theta function(x) x  
from.theta function(x) x
```

```
hyperid 29102
name theta2
short.name theta2
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
```

```
hyperid 29103
name theta3
short.name theta3
initial 1048576
fixed FALSE
```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta4
    hyperid 29104
    name theta4
    short.name theta4
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 29105
    name theta5
    short.name theta5
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 29106
    name theta6
    short.name theta6
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 29107
    name theta7
    short.name theta7
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta8

hyperid 29108
name theta8
short.name theta8
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta9

hyperid 29109
name theta9
short.name theta9
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta10

hyperid 29110
name theta10
short.name theta10
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta11

hyperid 29111
name theta11
short.name theta11
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta12

hyperid 29112
name theta12
short.name theta12

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta13
    hyperid 29113
    name theta13
    short.name theta13
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta14
    hyperid 29114
    name theta14
    short.name theta14
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta15
    hyperid 29115
    name theta15
    short.name theta15
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta16
    hyperid 29116
    name theta16
    short.name theta16
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta17
  hyperid 29117
  name theta17
  short.name theta17
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta18
  hyperid 29118
  name theta18
  short.name theta18
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta19
  hyperid 29119
  name theta19
  short.name theta19
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta20
  hyperid 29120
  name theta20
  short.name theta20
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta21
  hyperid 29121

```

```

    name theta21
    short.name theta21
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta22
    hyperid 29122
    name theta22
    short.name theta22
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta23
    hyperid 29123
    name theta23
    short.name theta23
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta24
    hyperid 29124
    name theta24
    short.name theta24
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta25
    hyperid 29125
    name theta25
    short.name theta25
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta26
    hyperid 29126
    name theta26
    short.name theta26
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta27
    hyperid 29127
    name theta27
    short.name theta27
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta28
    hyperid 29128
    name theta28
    short.name theta28
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta29
    hyperid 29129
    name theta29
    short.name theta29
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```


theta30

hyperid 29130
name theta30
short.name theta30
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta31

hyperid 29131
name theta31
short.name theta31
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta32

hyperid 29132
name theta32
short.name theta32
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta33

hyperid 29133
name theta33
short.name theta33
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta34

hyperid 29134
name theta34
short.name theta34

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta35
    hyperid 29135
    name theta35
    short.name theta35
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta36
    hyperid 29136
    name theta36
    short.name theta36
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta37
    hyperid 29137
    name theta37
    short.name theta37
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta38
    hyperid 29138
    name theta38
    short.name theta38
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta39
  hyperid 29139
  name theta39
  short.name theta39
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta40
  hyperid 29140
  name theta40
  short.name theta40
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta41
  hyperid 29141
  name theta41
  short.name theta41
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta42
  hyperid 29142
  name theta42
  short.name theta42
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta43
  hyperid 29143

```

```

name theta43
short.name theta43
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta44
hyperid 29144
name theta44
short.name theta44
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta45
hyperid 29145
name theta45
short.name theta45
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta46
hyperid 29146
name theta46
short.name theta46
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta47
hyperid 29147
name theta47
short.name theta47
initial 1048576
fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta48
    hyperid 29148
    name theta48
    short.name theta48
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta49
    hyperid 29149
    name theta49
    short.name theta49
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta50
    hyperid 29150
    name theta50
    short.name theta50
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta51
    hyperid 29151
    name theta51
    short.name theta51
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta52

hyperid 29152
name theta52
short.name theta52
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta53

hyperid 29153
name theta53
short.name theta53
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta54

hyperid 29154
name theta54
short.name theta54
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta55

hyperid 29155
name theta55
short.name theta55
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta56

hyperid 29156
name theta56
short.name theta56

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta57
    hyperid 29157
    name theta57
    short.name theta57
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta58
    hyperid 29158
    name theta58
    short.name theta58
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta59
    hyperid 29159
    name theta59
    short.name theta59
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta60
    hyperid 29160
    name theta60
    short.name theta60
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta61
  hyperid 29161
  name theta61
  short.name theta61
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta62
  hyperid 29162
  name theta62
  short.name theta62
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta63
  hyperid 29163
  name theta63
  short.name theta63
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta64
  hyperid 29164
  name theta64
  short.name theta64
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta65
  hyperid 29165

```



```

    name theta65
    short.name theta65
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta66
    hyperid 29166
    name theta66
    short.name theta66
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta67
    hyperid 29167
    name theta67
    short.name theta67
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta68
    hyperid 29168
    name theta68
    short.name theta68
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta69
    hyperid 29169
    name theta69
    short.name theta69
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta70
    hyperid 29170
    name theta70
    short.name theta70
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta71
    hyperid 29171
    name theta71
    short.name theta71
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta72
    hyperid 29172
    name theta72
    short.name theta72
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta73
    hyperid 29173
    name theta73
    short.name theta73
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta74

hyperid 29174
name theta74
short.name theta74
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta75

hyperid 29175
name theta75
short.name theta75
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta76

hyperid 29176
name theta76
short.name theta76
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta77

hyperid 29177
name theta77
short.name theta77
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta78

hyperid 29178
name theta78
short.name theta78

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta79
    hyperid 29179
    name theta79
    short.name theta79
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta80
    hyperid 29180
    name theta80
    short.name theta80
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta81
    hyperid 29181
    name theta81
    short.name theta81
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta82
    hyperid 29182
    name theta82
    short.name theta82
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta83
  hyperid 29183
  name theta83
  short.name theta83
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta84
  hyperid 29184
  name theta84
  short.name theta84
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta85
  hyperid 29185
  name theta85
  short.name theta85
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta86
  hyperid 29186
  name theta86
  short.name theta86
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta87
  hyperid 29187

```

```

    name theta87
    short.name theta87
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta88
    hyperid 29188
    name theta88
    short.name theta88
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta89
    hyperid 29189
    name theta89
    short.name theta89
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta90
    hyperid 29190
    name theta90
    short.name theta90
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta91
    hyperid 29191
    name theta91
    short.name theta91
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta92
    hyperid 29192
    name theta92
    short.name theta92
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta93
    hyperid 29193
    name theta93
    short.name theta93
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta94
    hyperid 29194
    name theta94
    short.name theta94
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta95
    hyperid 29195
    name theta95
    short.name theta95
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta96

hyperid 29196
name theta96
short.name theta96
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta97

hyperid 29197
name theta97
short.name theta97
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta98

hyperid 29198
name theta98
short.name theta98
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta99

hyperid 29199
name theta99
short.name theta99
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta100

hyperid 29200
name theta100
short.name theta100


```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta101
    hyperid 29201
    name theta101
    short.name theta101
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta102
    hyperid 29202
    name theta102
    short.name theta102
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta103
    hyperid 29203
    name theta103
    short.name theta103
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta104
    hyperid 29204
    name theta104
    short.name theta104
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta105
  hyperid 29205
  name theta105
  short.name theta105
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta106
  hyperid 29206
  name theta106
  short.name theta106
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta107
  hyperid 29207
  name theta107
  short.name theta107
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta108
  hyperid 29208
  name theta108
  short.name theta108
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta109
  hyperid 29209

```

```

name theta109
short.name theta109
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta110
hyperid 29210
name theta110
short.name theta110
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta111
hyperid 29211
name theta111
short.name theta111
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta112
hyperid 29212
name theta112
short.name theta112
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta113
hyperid 29213
name theta113
short.name theta113
initial 1048576
fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta114
    hyperid 29214
    name theta114
    short.name theta114
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta115
    hyperid 29215
    name theta115
    short.name theta115
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta116
    hyperid 29216
    name theta116
    short.name theta116
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta117
    hyperid 29217
    name theta117
    short.name theta117
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta118
hyperid 29218
name theta118
short.name theta118
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta119
hyperid 29219
name theta119
short.name theta119
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta120
hyperid 29220
name theta120
short.name theta120
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta121
hyperid 29221
name theta121
short.name theta121
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta122
hyperid 29222
name theta122
short.name theta122

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta123
    hyperid 29223
    name theta123
    short.name theta123
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta124
    hyperid 29224
    name theta124
    short.name theta124
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta125
    hyperid 29225
    name theta125
    short.name theta125
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta126
    hyperid 29226
    name theta126
    short.name theta126
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta127
  hyperid 29227
  name theta127
  short.name theta127
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta128
  hyperid 29228
  name theta128
  short.name theta128
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta129
  hyperid 29229
  name theta129
  short.name theta129
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta130
  hyperid 29230
  name theta130
  short.name theta130
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta131
  hyperid 29231

```

```

    name theta131
    short.name theta131
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta132
    hyperid 29232
    name theta132
    short.name theta132
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta133
    hyperid 29233
    name theta133
    short.name theta133
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta134
    hyperid 29234
    name theta134
    short.name theta134
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta135
    hyperid 29235
    name theta135
    short.name theta135
    initial 1048576
    fixed FALSE

```



```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta136
    hyperid 29236
    name theta136
    short.name theta136
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta137
    hyperid 29237
    name theta137
    short.name theta137
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta138
    hyperid 29238
    name theta138
    short.name theta138
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta139
    hyperid 29239
    name theta139
    short.name theta139
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta140

hyperid 29240
name theta140
short.name theta140
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta141

hyperid 29241
name theta141
short.name theta141
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta142

hyperid 29242
name theta142
short.name theta142
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta143

hyperid 29243
name theta143
short.name theta143
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta144

hyperid 29244
name theta144
short.name theta144

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta145
    hyperid 29245
    name theta145
    short.name theta145
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta146
    hyperid 29246
    name theta146
    short.name theta146
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta147
    hyperid 29247
    name theta147
    short.name theta147
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta148
    hyperid 29248
    name theta148
    short.name theta148
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta149
  hyperid 29249
  name theta149
  short.name theta149
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta150
  hyperid 29250
  name theta150
  short.name theta150
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta151
  hyperid 29251
  name theta151
  short.name theta151
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta152
  hyperid 29252
  name theta152
  short.name theta152
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta153
  hyperid 29253

```

```

    name theta153
    short.name theta153
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta154
    hyperid 29254
    name theta154
    short.name theta154
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta155
    hyperid 29255
    name theta155
    short.name theta155
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta156
    hyperid 29256
    name theta156
    short.name theta156
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta157
    hyperid 29257
    name theta157
    short.name theta157
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta158
    hyperid 29258
    name theta158
    short.name theta158
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta159
    hyperid 29259
    name theta159
    short.name theta159
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta160
    hyperid 29260
    name theta160
    short.name theta160
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta161
    hyperid 29261
    name theta161
    short.name theta161
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta162

hyperid 29262
name theta162
short.name theta162
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta163

hyperid 29263
name theta163
short.name theta163
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta164

hyperid 29264
name theta164
short.name theta164
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta165

hyperid 29265
name theta165
short.name theta165
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta166

hyperid 29266
name theta166
short.name theta166

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta167
    hyperid 29267
    name theta167
    short.name theta167
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta168
    hyperid 29268
    name theta168
    short.name theta168
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta169
    hyperid 29269
    name theta169
    short.name theta169
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta170
    hyperid 29270
    name theta170
    short.name theta170
    initial 1048576
    fixed FALSE
    prior none
    param

```



```

    to.theta function(x) x
    from.theta function(x) x
theta171
  hyperid 29271
  name theta171
  short.name theta171
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta172
  hyperid 29272
  name theta172
  short.name theta172
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta173
  hyperid 29273
  name theta173
  short.name theta173
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta174
  hyperid 29274
  name theta174
  short.name theta174
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta175
  hyperid 29275

```

```

    name theta175
    short.name theta175
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta176
    hyperid 29276
    name theta176
    short.name theta176
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta177
    hyperid 29277
    name theta177
    short.name theta177
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta178
    hyperid 29278
    name theta178
    short.name theta178
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta179
    hyperid 29279
    name theta179
    short.name theta179
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta180
    hyperid 29280
    name theta180
    short.name theta180
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta181
    hyperid 29281
    name theta181
    short.name theta181
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta182
    hyperid 29282
    name theta182
    short.name theta182
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta183
    hyperid 29283
    name theta183
    short.name theta183
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

```

theta184
  hyperid 29284
  name theta184
  short.name theta184
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta185
  hyperid 29285
  name theta185
  short.name theta185
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta186
  hyperid 29286
  name theta186
  short.name theta186
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta187
  hyperid 29287
  name theta187
  short.name theta187
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta188
  hyperid 29288
  name theta188
  short.name theta188

```

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta189
    hyperid 29289
    name theta189
    short.name theta189
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta190
    hyperid 29290
    name theta190
    short.name theta190
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta191
    hyperid 29291
    name theta191
    short.name theta191
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta192
    hyperid 29292
    name theta192
    short.name theta192
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta193
  hyperid 29293
  name theta193
  short.name theta193
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta194
  hyperid 29294
  name theta194
  short.name theta194
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta195
  hyperid 29295
  name theta195
  short.name theta195
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta196
  hyperid 29296
  name theta196
  short.name theta196
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta197
  hyperid 29297

```

```

    name theta197
    short.name theta197
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta198
    hyperid 29298
    name theta198
    short.name theta198
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta199
    hyperid 29299
    name theta199
    short.name theta199
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta200
    hyperid 29300
    name theta200
    short.name theta200
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta201
    hyperid 29301
    name theta201
    short.name theta201
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta202
    hyperid 29302
    name theta202
    short.name theta202
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta203
    hyperid 29303
    name theta203
    short.name theta203
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta204
    hyperid 29304
    name theta204
    short.name theta204
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta205
    hyperid 29305
    name theta205
    short.name theta205
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```


theta206

hyperid 29306
name theta206
short.name theta206
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta207

hyperid 29307
name theta207
short.name theta207
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta208

hyperid 29308
name theta208
short.name theta208
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta209

hyperid 29309
name theta209
short.name theta209
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta210

hyperid 29310
name theta210
short.name theta210

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta211
    hyperid 29311
    name theta211
    short.name theta211
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta212
    hyperid 29312
    name theta212
    short.name theta212
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta213
    hyperid 29313
    name theta213
    short.name theta213
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta214
    hyperid 29314
    name theta214
    short.name theta214
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta215
  hyperid 29315
  name theta215
  short.name theta215
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta216
  hyperid 29316
  name theta216
  short.name theta216
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta217
  hyperid 29317
  name theta217
  short.name theta217
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta218
  hyperid 29318
  name theta218
  short.name theta218
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta219
  hyperid 29319

```

```

    name theta219
    short.name theta219
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta220
    hyperid 29320
    name theta220
    short.name theta220
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta221
    hyperid 29321
    name theta221
    short.name theta221
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta222
    hyperid 29322
    name theta222
    short.name theta222
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta223
    hyperid 29323
    name theta223
    short.name theta223
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta224
    hyperid 29324
    name theta224
    short.name theta224
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta225
    hyperid 29325
    name theta225
    short.name theta225
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta226
    hyperid 29326
    name theta226
    short.name theta226
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta227
    hyperid 29327
    name theta227
    short.name theta227
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta228

hyperid 29328
name theta228
short.name theta228
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta229

hyperid 29329
name theta229
short.name theta229
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta230

hyperid 29330
name theta230
short.name theta230
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta231

hyperid 29331
name theta231
short.name theta231
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta232

hyperid 29332
name theta232
short.name theta232

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta233
    hyperid 29333
    name theta233
    short.name theta233
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta234
    hyperid 29334
    name theta234
    short.name theta234
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta235
    hyperid 29335
    name theta235
    short.name theta235
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta236
    hyperid 29336
    name theta236
    short.name theta236
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta237
  hyperid 29337
  name theta237
  short.name theta237
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta238
  hyperid 29338
  name theta238
  short.name theta238
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta239
  hyperid 29339
  name theta239
  short.name theta239
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta240
  hyperid 29340
  name theta240
  short.name theta240
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta241
  hyperid 29341

```



```

    name theta241
    short.name theta241
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta242
    hyperid 29342
    name theta242
    short.name theta242
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta243
    hyperid 29343
    name theta243
    short.name theta243
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta244
    hyperid 29344
    name theta244
    short.name theta244
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta245
    hyperid 29345
    name theta245
    short.name theta245
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta246
    hyperid 29346
    name theta246
    short.name theta246
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta247
    hyperid 29347
    name theta247
    short.name theta247
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta248
    hyperid 29348
    name theta248
    short.name theta248
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta249
    hyperid 29349
    name theta249
    short.name theta249
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta250

hyperid 29350
name theta250
short.name theta250
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta251

hyperid 29351
name theta251
short.name theta251
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta252

hyperid 29352
name theta252
short.name theta252
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta253

hyperid 29353
name theta253
short.name theta253
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta254

hyperid 29354
name theta254
short.name theta254

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta255
    hyperid 29355
    name theta255
    short.name theta255
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta256
    hyperid 29356
    name theta256
    short.name theta256
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta257
    hyperid 29357
    name theta257
    short.name theta257
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta258
    hyperid 29358
    name theta258
    short.name theta258
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta259
  hyperid 29359
  name theta259
  short.name theta259
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta260
  hyperid 29360
  name theta260
  short.name theta260
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta261
  hyperid 29361
  name theta261
  short.name theta261
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta262
  hyperid 29362
  name theta262
  short.name theta262
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta263
  hyperid 29363

```

```

    name theta263
    short.name theta263
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta264
    hyperid 29364
    name theta264
    short.name theta264
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta265
    hyperid 29365
    name theta265
    short.name theta265
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta266
    hyperid 29366
    name theta266
    short.name theta266
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta267
    hyperid 29367
    name theta267
    short.name theta267
    initial 1048576
    fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta268
    hyperid 29368
    name theta268
    short.name theta268
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta269
    hyperid 29369
    name theta269
    short.name theta269
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta270
    hyperid 29370
    name theta270
    short.name theta270
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta271
    hyperid 29371
    name theta271
    short.name theta271
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta272

hyperid 29372
name theta272
short.name theta272
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta273

hyperid 29373
name theta273
short.name theta273
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta274

hyperid 29374
name theta274
short.name theta274
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta275

hyperid 29375
name theta275
short.name theta275
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta276

hyperid 29376
name theta276
short.name theta276


```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta277
    hyperid 29377
    name theta277
    short.name theta277
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta278
    hyperid 29378
    name theta278
    short.name theta278
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta279
    hyperid 29379
    name theta279
    short.name theta279
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta280
    hyperid 29380
    name theta280
    short.name theta280
    initial 1048576
    fixed FALSE
    prior none
    param

```

```

    to.theta function(x) x
    from.theta function(x) x
theta281
  hyperid 29381
  name theta281
  short.name theta281
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta282
  hyperid 29382
  name theta282
  short.name theta282
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta283
  hyperid 29383
  name theta283
  short.name theta283
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta284
  hyperid 29384
  name theta284
  short.name theta284
  initial 1048576
  fixed FALSE
  prior none
  param
  to.theta function(x) x
  from.theta function(x) x
theta285
  hyperid 29385

```

```

name theta285
short.name theta285
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta286
hyperid 29386
name theta286
short.name theta286
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta287
hyperid 29387
name theta287
short.name theta287
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta288
hyperid 29388
name theta288
short.name theta288
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x
theta289
hyperid 29389
name theta289
short.name theta289
initial 1048576
fixed FALSE

```

```

    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta290
    hyperid 29390
    name theta290
    short.name theta290
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta291
    hyperid 29391
    name theta291
    short.name theta291
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta292
    hyperid 29392
    name theta292
    short.name theta292
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta293
    hyperid 29393
    name theta293
    short.name theta293
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

```

theta294

hyperid 29394
name theta294
short.name theta294
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta295

hyperid 29395
name theta295
short.name theta295
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta296

hyperid 29396
name theta296
short.name theta296
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta297

hyperid 29397
name theta297
short.name theta297
initial 1048576
fixed FALSE
prior none
param
to.theta function(x) x
from.theta function(x) x

theta298

hyperid 29398
name theta298
short.name theta298

```

    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta299
    hyperid 29399
    name theta299
    short.name theta299
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x
theta300
    hyperid 29400
    name theta300
    short.name theta300
    initial 1048576
    fixed FALSE
    prior none
    param
    to.theta function(x) x
    from.theta function(x) x

constr FALSE

nrow.ncol FALSE

augmented TRUE

aug.factor 1

aug.constr 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

n.div.by -1

n.required TRUE

set.default.values TRUE

pdf iidkd

```

Example

Just simulate some data and estimate the parameters back. This is for `order=4`.

```
library(mvtnorm)

n <- 300
m <- 4
N <- m*n
rho <- 0.8

Sigma <- matrix(NA, m, m)
diag(Sigma) <- (1/(1:m))^2
for(i in 1:m) {
  for (j in 1:m) {
    if (i != j) {
      Sigma[i, j] <- rho^abs(i-j) * sqrt(Sigma[i, i] * Sigma[j, j])
    }
  }
}

y <- c()
yy <- rmvnorm(n, sigma = Sigma)
for(i in 1:m) {
  y <- c(y, yy[, i])
}

r <- inla(y ~ f(i, model = "iidkd", order = m, n=N,
  ## set parameters using 'theta1'.
  ## these are the default parameters.
  hyper = list(theta1 = list(
    param = c(100, rep(1, m), rep(0, m*(m-1)/2)))),
  data = data.frame(i = 1:N, y),
  ## fix precision as we have exact observations
  control.family = list(hyper = list(
    prec = list(initial = 15, fixed = TRUE))),
  verbose = FALSE)

## this is how the internal parameters are defined
L <- t(chol(solve(Sigma)))
diag(L) <- log(diag(L))
LL <- t(chol(solve(cov(yy))))
diag(LL) <- log(diag(LL))

## compare the estimated (internal) parameters with MLE and the truth
round(dig = 3, cbind(true = c(diag(L), L[lower.tri(L)]),
  mle = c(diag(LL), LL[lower.tri(LL)]),
  inla = r$mode$theta))

## this gives a list of sampled matrices (stdev's and correlations)
xx <- inla.iidkd.sample(10^4, r, "i")
## compute the mean
qq <- matrix(rowMeans(matrix(unlist(xx), nrow = m^2)), m, m)

iSigma <- 1/sqrt(diag(Sigma))
Cor <- diag(iSigma) %*% Sigma %*% diag(iSigma)
round(dig = 3, cbind(inla = c(diag(qq), qq[lower.tri(qq)]),
  true = c(sqrt(diag(Sigma)), Cor[lower.tri(Cor)])))
```

Notes

Note that for higher values of k the only integration scheme that is supported is `control.inla=list(int.strategy="eb")`.